

Didier Merk

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Citizenship: Netherlands

Summary

- Master of Science in Artificial Intelligence from University of Amsterdam (*GPA: 8.2/10*).
- Former intern at **CERN** (Cloud Computing team) and **ING Bank** (Wholesale Banking Advanced Analytics team).
- *Strengths & stack*: Statistics and Probability Theory, Python, PyTorch, Spark/Azure; experience across time series analysis, fine-tuning LLM architectures for forecasting, deep learning, CV, and NLP; teaching experience as Graduate TA.

Education

University of Amsterdam

MSc in Artificial Intelligence

Sep. 2022 – Sep. 2024

- GPA 8.2/10; Thesis “Rethinking Models and Evaluations for Time-Series Forecasting”.
- Core focus on advanced Machine- and Deep Learning, Computer Vision, NLP and Information Retrieval.
- Graduate TA, Game Theory: co-led MSc tutorials on strategic interaction and equilibria.
- **Published research**: reproduced and extended *LASSI*, a deep-learning method for certified individual fairness, using generative deep neural network *GLOW*; open-sourced and accepted at NeurIPS workshop 2023.

BSc Major in Artificial Intelligence

Sep. 2020 – Jul. 2022

- Thesis “Hyperparameter Optimization for Jet Tagging” at the CMS experiment at CERN (grade 8.5).
- Focus on machine learning, logic, data structures & algorithms and mathematical foundation courses.

Experience

ING Bank

Machine Learning Intern

Amsterdam, NL

Jan. 2024 – Dec. 2024

- Researched the use of pre-trained transformers and LLM architectures in the domain of financial time-series forecasting for the Wholesale Banking Advanced Analytics team; engineered end-to-end forecasting pipelines.
- Findings resulted in master thesis, an **invited talk** at **ING’s DSCC 2024** and directly influenced the team’s research roadmap.

CERN

Research Scientist Intern

Geneva, CH

Mar. 2022 – Jul. 2022

- Designed a hyperparameter optimization study to improve the Particle Transformer model used to classify proton-proton collisions at the CMS detector.
- Worked for the Cloud Computing team; executed experiments on their novel Kubernetes’ Kubeflow platform and documented the improved model results in my bachelor thesis.

Publications

[Re] Study of “Latent Space Smoothing for Individually Fair Representations”

Rescience C

Didier Merk, Denny Smit, Boaz Beukers, Tsatsral Mendsuren - [10.5281/zenodo.8173725](https://zenodo.org/record/8173725)

2023

Skills

Algorithm Design: Design of end-to-end machine learning projects; large-scale data analysis; evaluation and statistical validation; optimization of machine learning models.

Coding: Python (PyTorch, NumPy, scikit-learn, neuralforecast), C, R, SQL, web development (HTML/CSS/JS); tooling: Git, Docker, Jupyter, LaTeX/Markdown

Math & Theory: Linear Algebra, Calculus, Probability Theory, (Bayesian) Statistics and Game Theory; foundations across Deep Learning, Computer Vision, NLP and Information Retrieval.

Projects

An Improvement on ‘Guiding Text-To-Image Diffusion Model Towards Grounded Generation’ github.com/didiermerk/grounded-diffusers

- Extended a grounded text-to-image diffusion system (Stable Diffusion) that outputs both an image and masks for the objects named in the prompt; built a plug-and-play pipeline with simple prompt design and regularized training, improving the mIoU score by **4%** using only **25%** of the data and **generalizing better** to unseen classes.
- Tools used: Python, PyTorch, HuggingFace Diffusers, CLIP, Stable Diffusion 1.5/2.0, CUDA (TITAN RTX).

Exploring Improvements for Medical Imaging on SegTHOR Dataset github.com/raoulritter/ai4mi

- Designed a reproducible pipeline for the segmentation of thoracic organs in CT scans. Improved the 3D-Dice score through preprocessing, data augmentation and model optimization by **18%** compared to the baseline ENet model.
- Tools used: Segmentation models (ENet, VM-Unet, SAM2); .NIfTI, 3D Slicer; Weights and Biases (wandb) sweeps.

Other activities

- Mentor a weekly class session, which helps secondary students facing educational and socioeconomic barriers with their homework, study habits and exam preparation and provides them with a warm meal.
- Fun fact: Performed at a sold out Royal Theater Carré in Amsterdam.
- Travel and sports enthusiast; football and fitness; outdoor sports climbing belay certified.
- Made a YouTube channel as a teenager with videos reaching tens of millions of views across all platforms.